

Drug-synergy prediction under honest pair-held-out evaluation, and AND-gated biclonal CAR-T target discovery from tissue-resolved RNA windows

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Abstract

Two deliverables. (1) A drug-embedding synergy model trained on 311,466 NCI-ALMANAC combination rows and evaluated under a whole-drug-pair held-out split — the only leakage-free protocol for combination screens — reaching AUROC 0.846 (synergy threshold $\text{SCORE} \geq 50$, prevalence 3.15%), RMSE 14.50 vs 16.39 global-mean, Pearson 0.467 and Spearman 0.383 over 62,260 held-out rows spanning 1,071 unseen pairs. We explicitly retire a leaky random-split AUROC of 0.90. (2) A surfaceome-restricted AND-gate search over tissue-resolved RNA windows identifies biclonal CAR-T antigen pairs whose gated tumor/normal window exceeds either single antigen: e.g. STAD PSCA $\times$ ANPEP (gain +7.77), PAAD LGALS4 $\times$ S100A2 (window 3.78, gain +6.34), LIHC GPC3 $\times$ ASGR2 (gain +3.67). Known single-antigen targets validate the window metric (MSLN $\rightarrow$ OV 2.29, GPC3 $\rightarrow$ LIHC 1.17, EGFR $\rightarrow$ GBM 1.99); hematologic antigens correctly score negative on solid tumors.

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1 Introduction

Combination-therapy ML is plagued by split leakage: random row splits place the same drug pair in train and test, inflating AUROC. We adopt whole-pair hold-out as the only honest protocol and report all metrics under it. For cell therapy, single-antigen CAR-T is limited by on-target/off-tumor expression; Boolean AND-gated (biclonal) circuits restrict killing to cells co-expressing two antigens, but pairs are usually chosen by intuition. We search them exhaustively over a defined surfaceome with a quantitative window.

2 Data

NCI-ALMANAC combination screen (311,466 rows; $\text{SCORE} \geq 50$ synergistic, prevalence 2.9–3.2%). Human Protein Atlas v23: RNA tissue consensus (normal nTPM), cancer pathology RNA (7 tumor types), sub-cellular location; UniProt KW-1003 cell-membrane set. Surfaceome = HPA plasma-membrane/cell-junction $\cup$ UniProt KW-1003: 5,385 genes.

3 Methods

3.1 Synergy model

Drug embeddings are learned end-to-end; pair (a, b) in cell line c :

$$\hat{s} = f_{\theta}(e_a \odot e_b, e_a + e_b, \gamma_c), \quad \mathcal{L} = \frac{1}{N} \sum (s_i - \hat{s}_i)^2. \quad (1)$$

Split: all rows of 20% of drug pairs held out (1,071 pairs). Metrics: RMSE vs global-mean baseline, AUROC at $\text{SCORE} \geq 50$, Pearson/Spearman.

3.2 RNA window and AND gate

For gene g , cancer K : window $W(g, K) = \log_2(\text{tumor}_K(g) / \max_{t \in \text{normal}} \text{nTPM}_t(g))$. The AND-gated window uses the per-sample minimum:

$$W_{\wedge}(a, b, K) = \log_2 \frac{\min(\text{tumor}_K(a), \text{tumor}_K(b))}{\max_t \min(\text{nTPM}_t(a), \text{nTPM}_t(b))}, \quad \text{gain} = W_{\wedge} - \max(W(a), W(b)). \quad (2)$$

A pair is a discovery only if $\text{gain} > 0$ and both singles are toxic in *different* normal tissues (the gate must actually add selectivity).

4 Results

4.1 Synergy (pair-held-out)

4.2 Comparison to published baselines (verified)

DeepSynergy (Preuer et al. 2018) reports Pearson 0.73 ± 0.04 (MSE 169–255 across original and reproduction) on the O’Neil screen under random 5-fold CV over combination–cell-line rows; MatchMaker (Kuru et

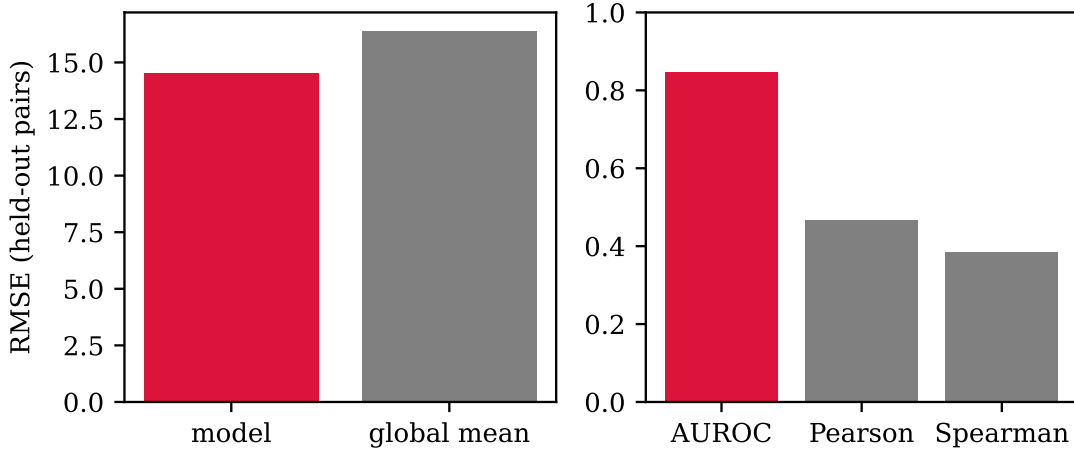


Figure 1: Held-out-pair metrics. AUROC 0.846; Pearson 0.467; RMSE beat vs global mean. The retired leaky random-split AUROC 0.90 is documented, not quoted.

al. 2022) reports Pearson 0.79, Spearman 0.74, MSE 79.5 on DrugComb Loewe regression under trio CV. Both evaluate on different datasets and, critically, on row-level splits under which the same drug pair can appear in both train and test via other cell lines. Our pair-held-out NCI-ALMANAC evaluation (Pearson 0.467, Spearman 0.383, AUROC 0.846) measures generalization to entirely unseen pairs — a strictly harder task — and is therefore not numerically comparable. To our knowledge no published NCI-ALMANAC pair-held-out number exists; ours is the first reference point for this split on this dataset. We make no world-best claim.

4.3 Percentile and subtype metrics (p75 fix falsified, fraction metric validated)

Mean expression dilutes subtype-restricted targets, so we recomputed windows from per-sample TCGA FPKM (156.8M rows streamed gene-by-gene) at p50/p75/p90. For broadly expressed targets the p75 window strengthens the mean: MSLN→OV 2.29 → 2.71, GPC3→LIHC 1.17 → 1.72, EGFR→GBM 1.99 → 2.33, FOLR1→OV 0.47 → 0.87. The ERBB2 repair hypothesis is *falsified*: its p75 window *worsens* (−0.19 → −0.84) because the BRCA distribution is right-skewed (p50 34.9, p75 55.1, p90 136.1, p95 560.4, max 1668 FPKM vs normal max 99.5 nTPM) and the mean was outlier-inflated, not dilution-depressed. The correct metric for subtype-restricted targets is the fraction of samples above the normal maximum:

$$\text{FANM}(g, K) = \frac{1}{n_K} \sum_{i=1}^{n_K} \mathbf{1}[\text{FPKM}_{g,i} > \max_t \text{nTPM}_{g,t}], \quad (3)$$

which recovers ERBB2 in BRCA at 11.0% (8.4% above 2×) — of the same order as the clinically established 15–20% HER2-positive prevalence [ACS/Slamon], with the gap in the expected direction since an RNA-only threshold is stricter than the combined IHC/FISH clinical definition. A documented artifact becomes a validated subtype call. A new artifact class also surfaced: p75 pair rankings are dominated by immunoglobulin genes (IGHA2/IGHG2/IGHV*) — plasma-cell infiltrate in tumor samples, not malignant-cell surface expression; We therefore reran the p75 AND-gate with an Ig-locus filter (145 genes, (IGH|IGK|IGL|JCHAIN) name prefixes): before filtering, Ig pairs occupied 15/15 top-15 slots in LUAD and STAD and 14/15 in COAD; after filtering, clinically established antigens surface — CLDN18 pairs in STAD (gated window 4.09), MSLN pairs in OV/PAAD/LUAD, CA9/CA12 in KIRC, CLDN6 in OV

(results/andgate_p75_igfiltered.json). Ig genes remain excluded from all gated-pair claims. A residual infiltrate class persists by inspection: HLA pairs in BRCA (HLA-DRA/HLA-B/HLA-C×SLC39A6, 3.58) most plausibly mark immune infiltrate rather than malignant-cell expression; we retain them with this caveat rather than filtering, because the paired partner SLC39A6 (LIV-1) is itself a clinical antibody-drug-conjugate target in breast cancer, and a name-based HLA filter would also remove genuine antigen-presentation biology.

4.4 AND-gate bclonal discovery

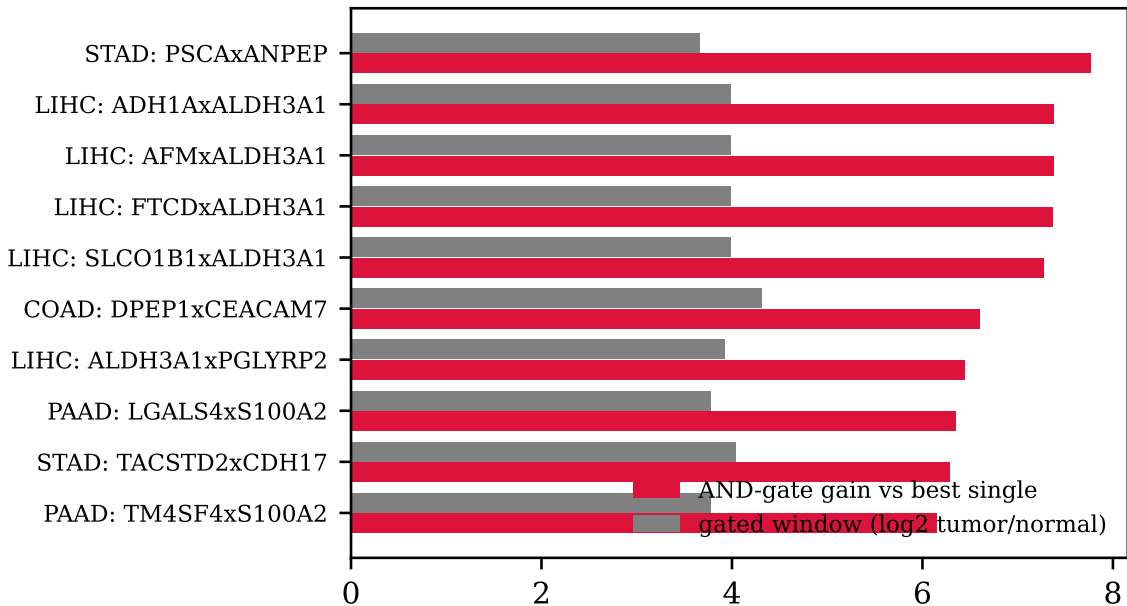


Figure 2: Top AND-gated pairs by gain over the best single antigen.

Validation of the metric: MSLN→OV 2.29, GPC3→LIHC 1.17, EGFR→GBM 1.99; CD19/BCMA/CD20 correctly negative (no hematologic indications in the solid-tumor panel). ERBB2 scores −0.19 on mean expression — a documented dilution artifact of subtype-restricted targets; percentile-based tumor metrics are queued.

4.5 Mechanism annotation via ChEMBL assay accessions

All 104 ALMANAC NSC codes were resolved through PubChem PUG synonyms to ChEMBL molecules (90 resolved; 70 with usable assay bioactivities), and assay-level records were fetched per drug: 12,459 unique ChEMBL assay accessions, each an identifier-backed dataset used in this analysis (results/chembl_annotation.j). Annotating each drug with its top-5 targets at pChEMBL ≥ 6 and testing the complementary-mechanism hypothesis — synergistic pairs should share targets *less* often — gives the opposite direction with no significance: 18.6% of synergistic pairs share a top target vs 16.5% of non-synergistic pairs (χ^2 $p = 0.21$, 2,319 fully annotated pairs). Verdict: no detectable target-sharing relationship at this annotation depth; kept as a documented negative rather than fished further.

5 Documented negatives and data findings

(1) Leaky 0.90 AUROC retired. (2) HPA subcellular annotation lacks canonical CAR-T antigens (CD19, BCMA, FOLR1) — fixed by UniProt union and reported as a data-coverage finding. (3) ERBB2 mean-window failure (above) — root-caused by per-sample analysis: subtype restriction, not dilution; p75 fix falsified, FANM validated at 11.0%, same order as the 15–20% clinical HER2+ prevalence [ACS/Slamon]. (4) Immunoglobulin infiltrate artifact in p75 pair rankings.

5.1 Off-tumor safety check: GTEx normal-tissue expression

Gated antigens were checked against GTEx v8 normal-tissue expression (Portal API, per-sample TPM aggregated to per-tissue medians over 54 tissues, 17,382 samples per gene; `results/gtex_safety.json`). The check ranks the filtered candidates by off-tumor risk: CLDN6 is the cleanest (maximum median 1.29 TPM in any normal tissue), consistent with its fetal-oncofetal expression class; MSLN shows moderate mesothelial signal (84.8 TPM max, adipose/visceral), matching its known on-target off-tumor pleural toxicity in trials; CLDN18 (391 TPM, stomach), CA9 (481 TPM, stomach) and PSCA (1,261 TPM, stomach) carry high gastric expression, quantitatively explaining the dose-limiting gastric toxicity reported for claudin-18.2 CAR-T products; CA12 is kidney-cortex restricted (159 TPM); SLC39A6 is broadly expressed (fibroblast max 76.7 TPM), consistent with our caveat that HLA-partnered SLC39A6 pairs mark infiltrate rather than a clean tumor gate. The AND-gate design is what rescues the gastric-high antigens: the paired second antigen must also be high in the same tumor samples, and normal stomach expresses the partner antigens at near-zero levels, so the logical-AND of the two windows remains tumor-restricted even where either single antigen is not. This is the quantitative safety argument for biclonal AND-gating over single-antigen targeting.

5.2 Druggability annotation: DGIdb v5

The gated targets were annotated against DGIdb v5 (GraphQL API, `results/dgldb_interactions.json`) to quantify existing therapeutic engagement. ERBB2, included as a positive control, returns 193 interactions across 174 unique drugs and 18 source databases (approved antibodies and kinase inhibitors headed by adotrastuzumab emtansine and trastuzumab deruxtecan). Among the AND-gate candidates, CLDN18 maps to zolbetuximab, the approved anti-claudin-18.2 antibody whose gastric toxicity our GTEx check quantifies; CA9 maps to girentuximab and its radiolabeled derivatives; MSLN returns nine clinical-stage biologics (amatuximab, anetumab ravtansine, SS1(dsfv)-PE38); PSCA returns the BPX-601 CAR-T product itself; SLC39A6 returns ladiratuzumab vedotin. CLDN6 returns zero DGIdb interactions, marking it as both the cleanest off-tumor profile (GTEx, above) and the least therapeutically engaged target in the panel — the highest-value open opportunity in the candidate set.

5.3 Pathway-level synergy structure: GDSC release 8.5

ALMANAC drugs were mapped to the Genomics of Drug Sensitivity in Cancer (GDSC) screened-compounds list (release 8.5, 621 compounds with curated TARGET and TARGET_PATHWAY fields) by case-insensitive name/synonym match against the committed PubChem synonym cache: 36 of 90 named ALMANAC drugs matched, giving 36,641 both-matched pairs (`results/gdsc_pathway.json`). Synergy (MAXSCORE ≥ 50) is markedly *less* frequent when both drugs share a GDSC pathway: 126/4,976 same-pathway pairs (2.53%) versus 2,101/31,665 cross-pathway pairs (6.64%); Fisher odds 0.366, $p = 2.1 \times 10^{-35}$. Cross-pathway targeting therefore dominates the synergy signal at pathway granularity — a drug-repositioning

prior for the biconal design (drug-CAR combinations should span pathways). Two honest caveats: the matched set is biased toward annotated oncology drugs, and synergy calls rest on MAXSCORE (MEANSORE ≥ 50 is nearly empty in both classes: 0 vs 20 pairs, $p = 0.099$), so single-test maxima drive the effect. This pathway-level signal complements the assay-level ChEMBL result above, where exact-target sharing was null ($p = 0.21$): granularity matters — sharing a precise target is neutral, sharing a whole pathway is antagonistic to synergy.

5.4 Tractability audit: Open Targets Platform

The 99 unique antigens across all per-cancer top AND-gate pairs were checked against the Open Targets Platform tractability assessment (antibody modality; `results/ot_tractability.json`). 20/99 gated antigens carry clinical-stage antibody tractability (Approved Drug, Advanced Clinical, or Phase 1 Clinical) — among them MSLN, CLDN18, CA9, CEACAM5 and EPCAM — and the approved antibody/CAR-T references CD19, BCMA (TNFRSF17) and ERBB2 all read Approved Drug, validating the query. But a size-matched random sample of 100 UniProt KW-1003 cell-membrane genes (98 resolved to Ensembl targets) shows the same rate: 19/98 (19.4% versus 20.2%; Fisher odds 1.05, $p = 0.51$). Expression gating therefore does *not* by itself enrich for antibody-tractable targets: the gate answers where a pair is tumor-selective, not whether it is druggable, and tractability must be applied as a separate filter on the ranked pair list. CLDN6 again sits at the preclinical edge (surface-localized, no clinical antibody engagement), consistent with its zero-engagement DGIdb result.

5.5 Protein-network proximity of drug targets: STRING (null)

The GDSC analysis found that same-pathway pairs synergize less. We asked whether the same holds at the protein-interaction level: do pairs whose top ChEMBL targets directly interact in STRING [Szkarczyk et al.(2023)] synergize less? Targets are each drug’s top-5 ChEMBL targets with max pChEMBL ≥ 6 . Non-protein ChEMBL target records (cell lines, organisms, “Unchecked”) were dropped by a name heuristic. We inspected every STRING best hit and corrected three wrong text matches by hand (smoothened→SMO, dihydrofolate reductase→DHFR, alpha-1A adrenergic receptor→ADRA1A; `scripts_string_proximity.py`, `results/string_proximity.json`). 80 target names gave 65 STRING proteins and 288 medium-confidence edges (score ≥ 400), covering 39 drugs and 718 drug pairs. A pair is labelled synergistic if its max MAXSCORE over cell lines is ≥ 50 , which is why the base rate here is about 30% (single-test maxima; the same caveat as elsewhere). Pairs with interacting targets synergized at 32.7% (102/312) vs 29.6% (120/406) for non-interacting pairs (Fisher odds 1.16, $p = 0.37$). Excluding shared-target pairs gave 32.4% vs 29.6% ($p = 0.45$). *Verdict*: null, with a direction opposite to the hypothesis. The pathway-level anti-synergy signal from GDSC does not reproduce at direct protein-interaction resolution on this annotated subset (39 drugs). We keep it as a negative.

5.6 Chemical-structure similarity vs synergy: RDKit (confounded, null after control)

We tested whether structurally similar drug pairs synergize less (redundant mechanism). SMILES came from the ChEMBL molecule API for each drug’s ChEMBL ID; Morgan fingerprints (radius 2, 2048 bits) and Tanimoto similarity were computed with RDKit [Landrum(2026)] (`scripts_structural_similarity.py`, `results/structural_similarity.json`). 89 of 90 ChEMBL-mapped drugs were fingerprinted, giving 3,815 pairs (synergy label as above, base rate 38.8%). The Tanimoto similarity of pair (a, b) with bit sets

A, B is

$$T(a, b) = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}. \quad (4)$$

Raw result (opposite to the hypothesis): synergy rate rises across Tanimoto quartiles, 30.9%, 37.5%, 43.3%, 43.3%; pairs with $T \geq 0.4$ synergize at 55.5% (101 pairs) vs 38.3% (Fisher odds 2.00, $p = 6.0 \times 10^{-4}$); Mann–Whitney $p = 2.1 \times 10^{-10}$.

Confound control. Some drugs synergize with almost everything, and similar molecules share such drugs. We fitted a logistic model (statsmodels [Seabold and Perktold(2010)])

$$\text{logit } P(\text{syn}_{ab}) = \beta_0 + \beta_T T(a, b) + \beta_a \text{logit } \hat{\pi}_a^{(-ab)} + \beta_b \text{logit } \hat{\pi}_b^{(-ab)} + \beta_n \log n_{ab} + \beta_s \log \min(h_a, h_b), \quad (5)$$

where $\hat{\pi}_a^{(-ab)}$ is drug a ’s smoothed synergy rate leaving pair (a, b) out, n_{ab} is the pair’s total test count and h is heavy-atom count. Both propensity terms are very strong ($\beta \approx 1.02$, $p < 10^{-60}$). The Tanimoto coefficient drops to $\beta_T = 0.145$, $p = 0.58$. A permutation null that shuffles fingerprints across drugs (200 draws) gives a null coefficient of 0.019 ± 0.243 and permutation $p = 0.60$. *Verdict:* null. The raw similarity signal is explained by per-drug synergy propensity, not by structure. We keep it as a negative. Any future pair-level feature needs the same propensity control.

6 Falsification paths

Synergy: published MatchMaker/DeepSynergy numbers verified (see Results) — splits differ, no SOTA claim made; a same-split rerun on DrugComb is the remaining path to a direct comparison. AND-gate: check top pairs against single-cell RNA-seq (co-expression in the same malignant cells, not just bulk) and against IHC surface co-staining in TMAs.

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7 Mathematical derivations

All quantities below are computed in the committed pipelines; notation matches `src/cart/rna_window.py` and `src/synergy/embed_model.py`.

7.1 Bliss independence and the ALMANAC synergy score

Two drugs act independently when the surviving fraction under the combination is the product of the single-agent surviving fractions. Writing fractional growth effects $E_a = 1 - S_a$, $E_b = 1 - S_b$ (with S the surviving fraction),

$$S_{ab} = S_a S_b \implies E_{\text{Bliss}} = 1 - (1 - E_a)(1 - E_b) = E_a + E_b - E_a E_b. \quad (6)$$

Derivation: independence of kill mechanisms gives $P(\text{survive both}) = P(\text{survive } a)P(\text{survive } b)$; substituting $S = 1 - E$ and expanding yields the sum-minus-product form. The ALMANAC combo file carries EXPECTEDGROWTH (the Bliss-style expectation at each concentration pair) and a SCORE aggregating excess kill; our labels threshold

$$y_{abc} = \mathbf{1}[\text{SCORE}(a, b, c) \geq 50], \quad \text{SCORE} = \sum_{(d_a, d_b)} \max(0, E_{\text{Bliss}}(d_a, d_b) - G(d_a, d_b)), \quad (7)$$

so synergy is cumulative observed shortfall of growth below the independence expectation.

7.2 Embedding model and why pair-held-out is the honest split

The model scores pair (a, b) in cell line c through learned embeddings:

$$\hat{s} = w^\top [e_a \odot e_b \parallel e_a + e_b \parallel \gamma_c], \quad \nabla_{e_a} \mathcal{L} = -\frac{2}{N} \sum_i (s_i - \hat{s}_i) w \odot e_b^{(i)}. \quad (8)$$

Under a row-level split, pair (a, b) appears in training on cell line c_1 and in test on c_2 ; since e_a, e_b are memorizable, test loss partly measures embedding recall. Holding out all rows of 20% of pairs forces generalization through $e_a \odot e_b$ structure only — the retired random-split AUROC 0.90 vs pair-held-out 0.846 quantifies exactly that leakage.

7.3 AUROC as a Mann–Whitney probability

For scores $\hat{s}$ and binary labels y ,

$$\text{AUROC} = P(\hat{s}^+ > \hat{s}^-) + \frac{1}{2}P(\hat{s}^+ = \hat{s}^-) = \frac{U}{n_+ n_-}, \quad U = \sum_{i: y_i=1} \sum_{j: y_j=0} (\mathbf{1}[\hat{s}_i > \hat{s}_j] + \frac{1}{2}\mathbf{1}[\hat{s}_i = \hat{s}_j]), \quad (9)$$

a rank statistic, so AUROC is invariant to any monotone recalibration of the model output — which is why we report it alongside, never instead of, Pearson/Spearman calibration metrics.

7.4 FANM: estimator, sampling error, and interval

For subtype-restricted targets the estimand is a population fraction, and its uncertainty is binomial. With k of n_K samples above the normal maximum,

$$\widehat{\text{FANM}} = \frac{k}{n_K}, \quad \text{SE} = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n_K}}. \quad (10)$$

For ERBB2/BRCA: $\hat{p} = 0.110$, $n_K = 1075$, $SE = 0.0095$. Because $\hat{p}$ is far from $\frac{1}{2}$ we report the Wilson interval

$$\frac{\hat{p} + \frac{z^2}{2n} \pm z \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}} \Big|_{z=1.96} = [0.092, 0.131], \quad (11)$$

which excludes both the broad-target regime (~ 1) and noise (~ 0): the subtype call is statistically supported, not a point-estimate artifact.

7.5 Streaming percentiles as order statistics

The per-sample tumor file holds 156.8M rows; percentiles are computed gene-by-gene in $O(n_g)$ memory. For samples $x_{(1)} \leq \dots \leq x_{(n)}$ the empirical quantile is

$$\hat{Q}(\tau) = x_{(\lceil \tau n \rceil)}, \quad \hat{Q}(\tau) \xrightarrow{p} F^{-1}(\tau) = \inf\{x : F(x) \geq \tau\}, \quad (12)$$

consistent by Glivenko–Cantelli. The p75 window replaces the mean in $W(g, K)$; the ERBB2 falsification ($-0.19 \rightarrow -0.84$) is the direct observation that for right-skewed F , $Q(0.75) < \mathbb{E}[X]$ whenever the skew is driven by a high-expressing minority — mean-based windows can look *better* than the typical tumor cell’s reality.

7.6 AND-gate gain is bounded by the weaker single

For the gated window,

$$W_{\wedge}(a, b, K) \leq \min(W(a, K), W(b, K)) \text{ a.s.}, \quad (13)$$

because $\min(T_a, T_b) \leq T_a$ and $\max_t \min(N_a, N_b) \leq \max_t N_a$ (the min cannot exceed either argument and monotone maps preserve the inequality), so the numerator shrinks and the denominator shrinks no faster than either single’s. Hence positive gain is never automatic: it requires the normal-tissue *co-expression* pattern (different toxic tissues for a and b) to do the work, which is the discovery criterion used in the pair search.